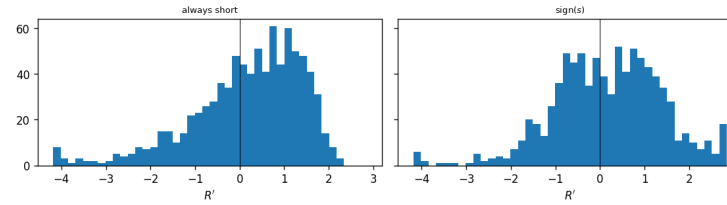


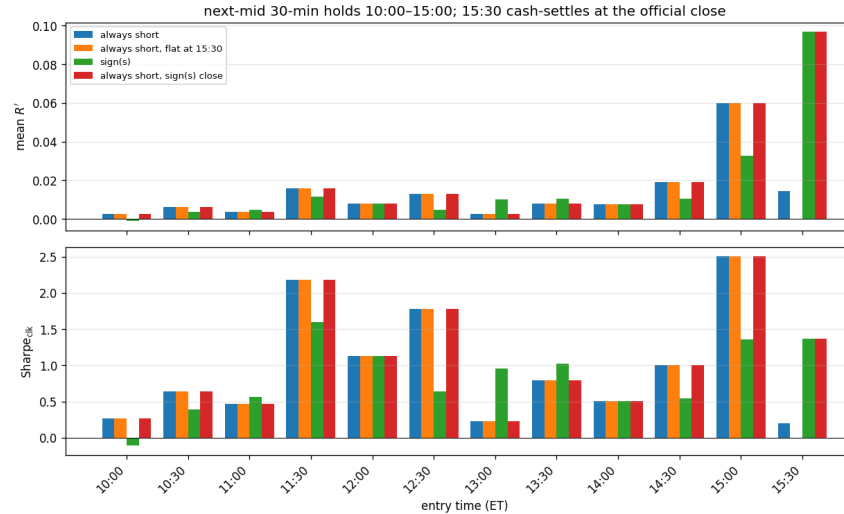
ODTE nearest-OTM package: per-rule return summary, pooled over the twelve entry windows 10:00–15:30 ET (daily sums), models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{ann}	n_{buy}	%buy	Sharpe _{ann} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.160	1.338	-11.40	2.24	-1.92	8.73	3.53	1.90	0	0.0	-1.92
<i>sign(s) (± 1 on the sign of s)</i>												
baseline (HAR + calendar OLS)	866	0.163	1.356	-5.96	11.70	0.27	7.40	3.54	1.91	266	30.7	-2.35
block-diagonal ridge	866	0.198	1.355	-6.26	11.40	0.55	7.39	4.30	2.32	252	29.1	-1.94
block-diagonal ridge, without the FOMC columns	866	0.192	1.348	-6.15	11.40	0.60	7.41	4.19	2.26	267	30.8	-2.02
LightGBM [†]	866	0.179	1.347	-5.96	11.40	0.58	7.25	3.92	2.12	277	32.0	-2.16
XGBoost [†]	866	0.173	1.357	-5.96	11.40	0.55	7.30	3.75	2.02	271	31.3	-2.21
lasso (causally tuned) [†]	866	0.180	1.337	-6.58	11.70	0.37	8.07	3.96	2.13	276	31.9	-2.17
lasso ($\alpha = 10^{-4}$)	866	0.209	1.344	-6.26	10.65	0.63	6.23	4.59	2.47	248	28.6	-1.82
elastic net (causally tuned) [†]	866	0.151	1.361	-6.72	11.40	0.31	7.19	3.26	1.76	266	30.7	-2.47

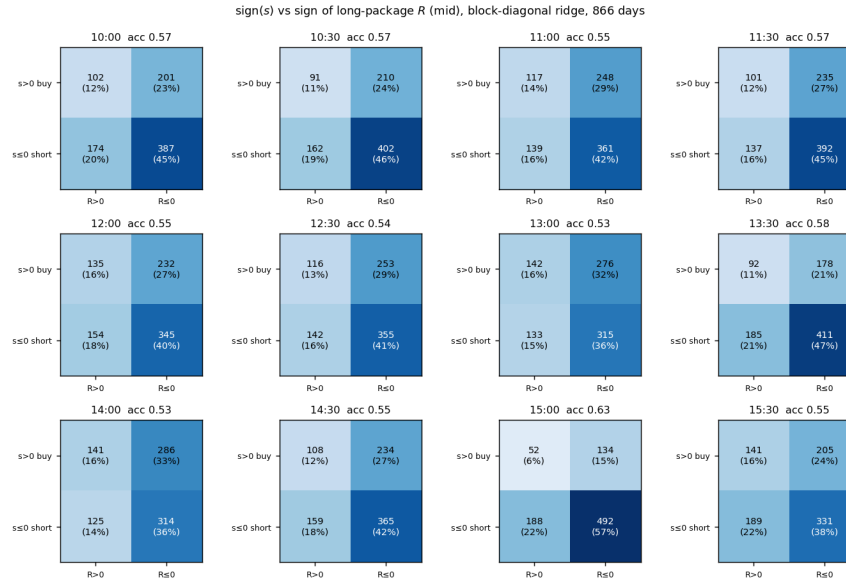
block-diagonal ridge, midpoint fills: daily sum over the twelve entry windows 10:00-15:30 ET; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



Distribution of the return of the daily sum over the twelve entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and Sharpe_{clk} by entry clock (grouped bars). Sharpe_{clk} is that clock's daily series $\times \sqrt{252}$, not the pooled Sharpe_{ann}. Next-mid 30-minute holds 10:00–15:00; 15:30 cash-settles at the official close.



Confusion matrix per entry clock: $\text{sign}(s)$ (buy if $s > 0$) against the sign of the long-package midpoint return. 15:30 is the paper trade.

Each row is the daily sum over the twelve entry windows 10:00–15:30 ET: one number per expiration day, then the usual statistics on that daily series. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of that clock’s series: one 30-minute hold per expiration day. It answers “if I only entered at this clock, what is my annual Sharpe?” It is *not* the book’s $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of $R_d^{\text{day}} = \sum_t R'_{d,t}$, the sum of the day’s twelve holds. That reserved name is the book’s daily P&L after pooling the session. At 15:30 the two coincide (one trade that day); they do not on 10:00–15:00 or on the pooled page. $\text{Sharpe}^\times$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); $\% \text{buy}$ is $100 \cdot n_{\text{buy}}/n$, the notebook’s column. The other column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^\times$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day’s remaining-session share of realized variance, $RV_t / \sum_{s \geq t} RV_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.